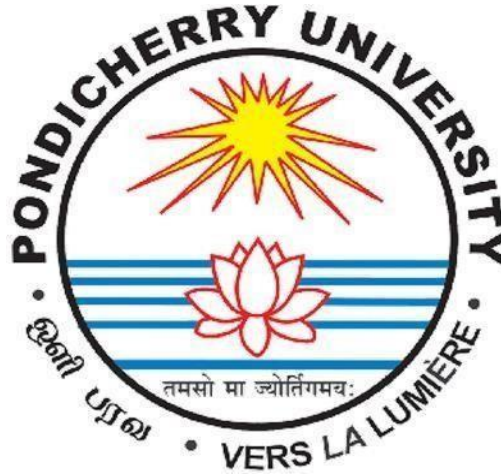


PONDICHERRY UNIVERSITY
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DEPARTMENT OF COMPUTER SCIENCE
MASTER OF COMPUTER APPLICATION
MAIN PROJECT REPORT
ZEROTH REVIEW

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TITLE OF THE PROJECT: SMARTFILE: SMART FILE DUPLICATE DETECTION AND STORAGE MANAGEMENT SYSTEM

UNDER THE GUIDENCE OF: Dr. S. L. JAYALAKSHMI

COURSE: MCA II – YEAR.

ABSTRACTION:

SmartFile is a web-based file duplicate detection and storage management system designed to help users identify duplicate files and manage their storage more efficiently. Duplicate copies of documents, images, downloads, and other files can accumulate over time and consume unnecessary storage space.

The proposed system allows users to scan a selected folder and identify files with identical content using file hashing. Files with matching hash values can be identified as duplicates even when they have different file names. SmartFile is intended to benefit individual computer users, students, office users, and organizations that manage a large number of files. The system aims to provide a simple and user-friendly way to identify duplicate files and understand unnecessary storage usage.

INTRODUCTION:

With the increasing use of computers, users store a large number of digital files such as documents, images, videos, downloads, assignments, and project files. Duplicate copies can be created unintentionally through repeated downloads, file sharing, backups, and manual copying. Over time, these duplicate files can occupy unnecessary storage space and make file management difficult.

Identifying duplicate files manually can be time-consuming, especially when a large number of files are stored. File names alone are not always sufficient because files with different names may contain the same content.

SmartFile is proposed to address this problem by allowing users to scan folders and identify duplicate files using file hashing. The system is useful for individuals, students, office users, and organizations that need to manage large numbers of files. It aims to provide a simple and user-friendly solution for identifying duplicate files, understanding storage usage, and managing files more efficiently.

SUMMARY:

1. Objective:

- To develop a system for detecting duplicate files and helping users manage unnecessary storage usage.

2. Key Steps:

- **Folder Scanning:** Scan files from a user-selected folder.
- **File Processing:** Collect and process file information.
- **Hash Generation:** Generate hash values for files based on their content.
- **Duplicate Detection:** Compare hash values to identify files with identical content.
- **Storage Analysis:** Analyze duplicate files to identify unnecessary storage usage.
- **Result Visualization:** Display scan information and duplicate file results to the user.
- **Scan History:** Store scan-related information for future reference.

3. Challenges:

- Handling different file types and a large number of files during folder scanning.
- Efficiently identifying duplicate files while processing file data.
- Ensuring proper communication between the frontend and backend.
- Addressed using file hashing, organized backend processing, database storage, and API-based communication.

PROJECT WORKFLOW:

The SmartFile system follows a simple workflow to scan files, identify duplicates, and display the results to the user.

1. **User Input** – The user enters the folder path that needs to be scanned through the React frontend. This folder path is used as the input for the duplicate file detection process.
2. **API Request** – The React frontend sends the entered folder path to the FastAPI backend using Axios. This allows the frontend to communicate with the backend and initiate the file scanning process.
3. **File Scanning** – After receiving the folder path, the FastAPI backend processes the request and uses Python to scan the files available in the selected folder. The files are then prepared for further duplicate detection.
4. **Hash Generation** – A unique hash value is generated for each scanned file based on its content. This hash value represents the actual content of the file and is used for comparison.
5. **Duplicate Identification** – The hash values of the scanned files are compared with each other. Files having the same hash value are identified as duplicate files, even if their file names are different.
6. **Data Storage** – The scan-related information is stored in the SQLite database. This allows the system to maintain the information generated during the scanning and duplicate detection process.
7. **Result Display** – After identifying the duplicate files, the results are sent back from the FastAPI backend to the React frontend. The duplicate files are then displayed to the user for review, helping them understand unnecessary storage usage.

This workflow helps SmartFile identify duplicate files efficiently and helps users understand unnecessary storage usage.

TOOLS AND TECHNOLOGIES USED:

1. React:

- Used to develop the frontend user interface of the SmartFile system

2. CSS:

- Used to design and style the user interface.

3. Axios:

- Used for communication between the frontend and backend.

4. Python:

- Used for file processing and backend logic.

5. FastAPI:

- Used to develop and handle the backend API.

6. Uvicorn:

- Used to run the FastAPI backend application.

7. SQLite:

- Used to store scan-related information and data.

8. File Hashing:

- Used to identify duplicate files by comparing their content.

FUTURE ENHANCEMENTS:

SmartFile will be further enhanced by improving the frontend and backend integration, grouping scanned files based on their formats such as images, documents, videos, audio, and other file types, and automatically detecting duplicate images during the normal folder scanning process. It will also provide more accurate storage usage and wasted storage information, options to safely manage or remove selected duplicate files, improved file filtering and sorting, and better organization of scan results. The system will also be extensively tested with different folder sizes and file types to improve its accuracy, performance, and reliability

CONCLUSION:

This project provides an efficient method for identifying duplicate files and managing unnecessary storage through automated folder scanning and file analysis. By automating key processes such as file scanning, file-format grouping, hash-based duplicate detection, and duplicate image identification, SmartFile helps users understand their storage usage and organize their files more effectively. The system provides a simple and user-friendly approach to finding unnecessary copies of files while reducing the effort required for manual file management. Its adaptability to future enhancements, such as improved duplicate detection, detailed storage analysis, and safe file management options, gives SmartFile the potential to become a more comprehensive and reliable solution for efficient digital file organization and storage management.